

RELIABILITY PAPER

Artificial intelligence and service performance: empirical findings from the hospitality sector

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Abstract

Purpose – This study aims to analyse the factors influencing AI adoption in the hospitality sector, utilising the technology acceptance model (TAM) and the technology-organisational-environmental (TOE) framework to understand how Industry 4.0 pressures drive the integration of emerging technologies, such as artificial intelligence (AI).

Design/methodology/approach – Data were collected via structured questionnaires from 312 valid responses from luxury hotels offering lodging and food services, and the hypotheses were tested using structural equation modelling.

Findings – This study demonstrates that perceived ease of use, competitive pressure, robust IT infrastructure and supportive government policies significantly drive AI adoption, thereby enhancing service performance. These factors also supplement each other: technological innovations enable new functionality, organisational preparation prepares for implementation and exogenous factors accelerate adoption processes, all together building significant momentum for the digital transformation of the hospitality industry.

Practical implications – The results give a hotel management, policy, and tourism stakeholder AI technology deployment strategic framework for boosting service performance, operation efficiency improvement and competitive advantage entrenchment during an industry being accelerated digitalised at a greater speed.

Originality/value – This study advances the literature by offering a novel, integrative perspective on AI adoption in the hospitality sector through the combined application of the TAM and the TOE framework. Positioned within a developing economy, it adopts a rigorous empirical lens to foreground organisational decision-making, thereby moving beyond the dominant customer-centric discourse. Further, by empirically linking AI adoption with service performance outcomes, the study bridges the critical gap between adoption intent and operational realisation, with particular relevance to the underexplored context of luxury hotels in India.

Keywords Artificial intelligence (AI), Technology acceptance model (TAM), Technology-organisational-environmental (TOE), Hospitality sector, Service performance

Paper type Research article



1. Introduction

The global tourism and hospitality industry is among the most dynamic and rapidly growing industries in the international service economy (Law *et al.*, 2022). This sector covers a

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diversified range of services such as accommodation, food and beverage, travel and entertainment that is deeply rooted in cultural, economic and environmental systems (Bhuiyan *et al.*, 2024). With rising pressure to provide high-quality, resource-efficient, and environmentally friendly services, tourism and hospitality businesses are increasingly adapting to the global sustainable development agenda (Ramaano, 2023; Saseanu *et al.*, 2020). At the same time, hospitality organisations are facing increased competitive pressure (CP) due to globalisation, economic recession, pandemics, and the blistering development of customer expectations (Chen, 2023). Customer satisfaction is a key factor in service effectiveness in the hospitality sector; however, inconsistencies are common in the gap between expected and actual service delivery (Rini and Kusumawardhani, 2024; Sharma *et al.*, 2022). To address these challenges, hospitality companies are increasingly adopting advanced digital technologies to enhance service stability, efficiency, and responsiveness, as well as to improve long-term business expansion (Ramaano, 2025a; Talwar *et al.*, 2022). One of these technologies is artificial intelligence (AI), which has been a transformative force, having the ability to significantly restructure the service delivery, decision-making processes, and the overall customer experience.

Chatbots, predictive analytics, recommendations, and service robots are AI-based technologies increasingly used to automate service processes, resource utilisation, and customer interactions (Saydam *et al.*, 2022; Tung and Au, 2018). Although it has potential, the use of AI in the hospitality industry is still skewed and problematic, raising some fundamental questions about what factors drive adoption choices and how much AI can be used to deliver improved service performance (SP).

The operational efficiency has increased significantly and the service customisation has become more accessible to hospitality firms due to technologies, including cloud computing, big data analytics, the Internet of Things (IoT), and robotics (Flavián *et al.*, 2022; Law *et al.*, 2022). As shown by the empirical data in the United States and Europe, the use of AI-based applications enhances customer experience and efficiencies in the back-end (Belanche *et al.*, 2021; Tung and Au, 2018). Similarly, East Asian-based hospitality companies such as Japan, China, and South Korea have been the first to implement AI by introducing robots as concierges, facial recognition systems, and intelligent booking systems (Lin *et al.*, 2020; Li *et al.*, 2025). Research based in the Middle East and the Gulf area also highlights the impact of AI as an inseparable part of the overall digital transformation policies due to the competitive forces and changing consumer behaviour (Nam *et al.*, 2021; Chen *et al.*, 2022).

Although such global trends have taken place, the scholarly research concerning the field of hospitality still presents considerable gaps in theoretical and empirical knowledge. The literature of the field has so far focused mainly on customer attitudes and intentions towards AI-based services (Lin *et al.*, 2020; Fusté-Forné, 2021; Samala *et al.*, 2020), and the mechanisms of organisational and managerial decision-making are underresearched. This omission is especially consequential to the hospitality industry in which the management of technological adoption is conditional on the strategic alignment and the performance and organisational readiness of organisations. In addition, the majority of empirical studies have been done in developed economies, thus limiting the extrapolation of results to emerging markets, where institutional factors, technological infrastructures, as well as organisational capabilities differ significantly (Bhuiyan *et al.*, 2024).

The hospitality industry in developing nations like India is growing at a very high pace and is forecasted to expand at a rate of 10.35 0.35 per annum by 2028 (IBEF, 2021). Already international hotel chains are expanding, and are projected to take 50% of the industry by 2024, versus 47% in 2022. Nevertheless, the existing empirical studies on the implementation of AI in the specified scenario are scarce (Kumar *et al.*, 2024; Nahar, 2024; Srivastava and Pal, 2024). Theoretically, hospitality research has mostly been based on either the Technology Acceptance Model (TAM) in isolation or the Technology-Organisation- Environment (TOE) model in isolation only, providing some partial explanations of AI-adoption behaviour. The combination of TAM and TOE to study the adoption of AI and its performance

implications is a research area that has not been thoroughly investigated. This is the limitation that does not allow a comprehensive understanding of the joint effect of technological, organisational, and environmental factors on AI adoption and SP.

In order to fill these identified gaps, the current study aims at exploring the determinants of AI adoption in hospitality industry and to evaluate the resultant impacts of artificial intelligence adoption on service delivery using a composite TAM-TOE framework. The study has taken a managerial approach, acknowledging that the adoption of AI is mainly defined by organisational decision-making and not by consumer acceptance. With the incorporation of technology perceptions, organisational preparedness as well as environmental pressures into a unified model, the study will shed some light on the factors that influence adoption intentions and subsequent effect on SP.

This study is structured around three interrelated research objectives. *First*, it investigates the influence of key technological determinants—perceived ease of use (PEOU) and perceived usefulness (PU)—on managerial intention to adopt AI within the hospitality sector. *Second*, it examines the role of organisational and environmental contingencies, specifically top management support (TMS), IT infrastructure (ITI), government support (GS), and competitive pressure (CP), in shaping managerial adoption intentions. *Third*, it assesses the downstream impact of AI adoption intention on service performance outcomes, namely efficiency, responsiveness, and customer satisfaction.

By integrating the TAM and TOE frameworks, the study develops a comprehensive analytical model that captures the interplay of technological, organisational, and environmental factors in influencing AI adoption decisions, while further evaluating how such adoption translates into measurable service performance outcomes.

- RQ1. What are the effects of technological factors (PEOU and PU), organisational factors (TMS and ITI) and environmental factors (GS and CP) on the intention of managers to use AI in the hospitality industry?
- RQ2. What is the effect of the intention to adopt AI on the performance outcomes of service in terms of efficiency, responsiveness, and customer satisfaction?

By answering these questions through empirical research into luxury hotels in India, the study will provide contextually based information on the adoption of AI in the emerging economy. This study makes a number of important contributions to the research on technology adoption, hospitality management, and quality and reliability management. To begin with, it contributes to theory because it proposes the incorporation of the TAM and the TOE framework into one conceptual framework to address the cumulative impact of technology, organisational and environmental factors on the adoption of AI. This combines the shortcomings of previous studies that assumed single paradigm models of study and provides a more comprehensive explanation of the adoption of organisational technology.

Second, the research is empirical and makes a shift in the analysis, where the focus is on the customers and is shifted to the managers. Though it has been observed in previous studies that customer intentions towards AI-enabled services were explored (Fusté-Forné, 2021; Lin *et al.*, 2020), little research has been conducted on how managers intend to adopt AI to improve customer service. This research will be instrumental in bridging an important gap in the literature on hospitality and enhancing the comprehension of the process of making a decision related to the adoption of AI within organisations by analysing the perceptions of managers.

Third, the article expands the research on AI adoption to the setting of a developing economy. The study answers researchers who argue that studies on the technology adoption process should come with more contextual diversity and strengthen the external validity of pre-existing theoretical frameworks by paying attention to the hospitality industry in India. Methodologically, the study uses structural equation modelling that tests hypothesised relationships, thus filling the gap of lack of quantitative, hypothesis-driven studies in this area.

Lastly, the adoption of AI as dependent on SP brings empirical information on how the digital technologies can improve the quality of service, operational efficiency, and customer satisfaction, thereby contributing to research on quality and reliability management.

The rest of this paper is organised in the following way. [Section 2](#) is a survey of the available literature on AI adoption, technology acceptance, and SP in the hospitality industry, and then creates the conceptual framework and hypotheses. [Section 3](#) highlights the research methodology, which includes the survey design, sampling protocols, data collection procedures and methods of data analysis. The empirical findings are provided in [Section 4](#), which includes a description of the measurement validation and results of the structural equations modelling. [Section 5](#) explains what the findings entail in terms of the current scholarly knowledge, including theoretical contributions and managerial implications. [Section 6](#) summarises the main findings of the research, provides the limitations of the study, and defines research directions in the future.

2. Literature review and hypothesis development

2.1 AI adoption and SP

Information technology is causing a tremendous revolution in the hospitality industry as it reinvents the concept of design, service delivery and service evaluation. In such a business environment, as competition and uncertainty continue to grow, technological changes are more than crucial to tourist attractions, improved customer satisfaction and long-term growth (IBEF, 2021; Pizam *et al.*, 2024). The trend to become digital allows hospitality businesses to simplify operations, minimise expenses, optimise the use of resources, and personalise services, which creates an overall impact on the improved performance of the services (Mishra *et al.*, 2022; Olan *et al.*, 2021). Within this wider technological transformation, AI has become a critical factor in service innovation, operational efficiency and client-focused service design.

The AI technologies, such as chatbots, virtual assistants, service robots, voice-controlled devices, and predictive analytics, are being increasingly implemented in the work of the hospitality industry (Ozekici *et al.*, 2025; Bhuiyan *et al.*, 2024). These innovations enhance the quality of services, as they contribute to real-time responsiveness, providing guests with individually tailored recommendations and ensuring smooth interactions between the guest and the hotel. Additionally, they improve the performance of operations by automating operations, improving the accuracy of demand prediction, and managing staffing and resource allocation (Naz *et al.*, 2024). As a result, the operating costs of organisations that implement AI will decrease by a significant margin of up to 15%, besides noteworthy increases in revenue and efficiency of services delivered (Pinheiro *et al.*, 2021). According to industry projections, by 2030, robotisation and AI have a possibility to replace close to a quarter of the hospitality workforce and, at the same time, increase service productivity and customer experience (Law *et al.*, 2022). The COVID-19 crisis increased the interest in AI, as hospitality companies began searching for contactless technologies to boost their safety, customer trust, and profitability (Gössling *et al.*, 2020; Shin and Kang, 2020). The concept of AI-based automation targets the labour-related issues, as well as providing the continuity of the services in case of crisis, which consolidates the strategic value of the concept. Also, the major brand hotel chains, including Hilton, Marriott, and Uber, have already implemented AI to predict the preferences of consumers, streamline the process of booking, and provide very personal services (Alsetoohy *et al.*, 2019).

In spite of these developments, the empirical evidence on the connection between AI adoption and performance of the service is limited, especially in developing economies such as India (Bhushan, 2021; Goel *et al.*, 2022). A large portion of the available literature is focused on consumer acceptance, the policy implications, or theoretical frameworks (Chatterjee, 2020; Lin *et al.*, 2020; Talwar *et al.*, 2022), and the gap in knowledge about the transformation of managerial AI adoption decisions into improved SP remains significant. Filling this void is

2.2 Theoretical underpinning

This study uses the TAM and TOE framework. This study uses TAM as one of its models because understanding technology acceptance enables more accurate forecasting of new information resources. This study uses the TOE framework as one of its models because it explains the latest technological innovation adoption in organisations and how the TOE influences its adoption and implementation.

Davis (1989) developed the TAM to understand the intention to adopt any technology. In previous studies, TAM has been used to investigate individuals' acceptance towards various technologies. The TOE framework was developed to investigate the factors impacting behavioural intentions towards technology adoption. Presently, this study uses both the framework (TAM and TOE) to evaluate the acceptance of AI in hotels. It is compulsory to provide a valid explanation for the technology adoption where these two frameworks will best suit. TOE frameworks explain technology adoption using three constructs: technology, organisational, and environmental (Abdurrahman *et al.*, 2024). On the other hand, TAM uses two main variables: PEOU and PU (Burgess *et al.*, 2023). TAM-TOE integration is best for investigating the socio-environmental and technological factors that contribute to AI acceptance.

(Alsetoohy *et al.*, 2019) explored the attitude of managers in adopting intelligent agent technology in hotel food SC using the TOE framework. The hospitality sector is primarily focused on delivering exceptional customer service, and AI can fundamentally transform the guest experience. AI has the potential to enhance customer satisfaction, increase efficiency, and drive revenue through various means, such as personalised recommendations, dynamic pricing that responds to customer needs, and chatbots that assist guests with their requirements (Cai *et al.*, 2022). Hotels can automate and optimise these operations using AI-powered solutions, increasing efficiency and effectiveness.

2.3 Conceptual framework and hypothesis formulation

Employing the TAM (Davis, 1989) and the TOE framework (DePietro *et al.*, 1990), this study gives a conceptual framework for facilitating empirical examination of hospitality industry-level adoption of AI at the organisational level. This framework combines organisational and environmental-level factors and individual-level cognitive factors to shed light on organisational managers' IAAI adoption and its subsequent effects on SP. From a TAM perspective, PEOU and PU reflect managers' perceptions of AI technologies, which inform their adoption intentions. From a TOE perspective, GS and CP capture external environmental drivers, while TMS and ITI reflect internal organisational capability and preparedness. The six antecedents hypothesised here are expected to positively affect IAAI, which, in turn, should improve SP, as measured by service quality, responsiveness, operational efficiency, and customer satisfaction. The proposed conceptual model is shown in Figure 1. This depicts the relationships between AI adoption precursors, AI adoption intention, and the eventual effect on SP.

2.3.1 *Perceived ease of use (PEOU)*. The hypothesis formulation for this study is grounded on the TAM and TOE framework. PEOU means the person's self-confidence towards any technology that can be used with less effort (de Kervenoael *et al.*, 2020). Abou-Shouk *et al.* (2021) found a positive relationship between PEOU and PU for adopting robots in hotel agencies. Users who intend to use the technology need to feel motivated by knowing the benefits. PEOU also includes the concepts of anxiety, enjoyment, playfulness, awareness of external control, and self-efficacy (Venkatesh and Bala, 2008). Seo and Lee (2021) studied the adoption of AI-based service robots in restaurants and found a positive association between PEOU and PU. Pai *et al.* (2018) analysed Chinese tourists' responses towards biometric

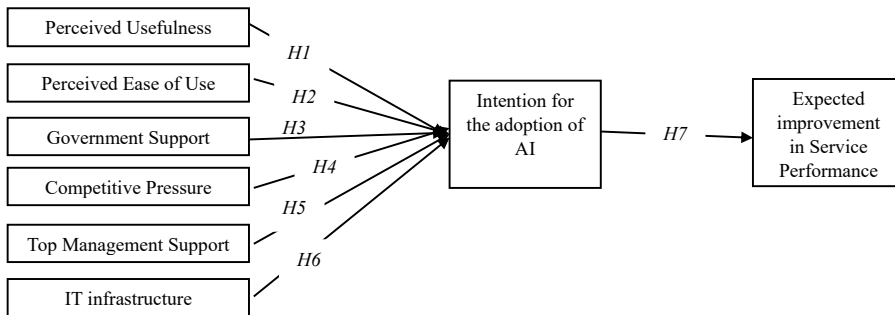


Figure 1. Theoretical model for the intention to adopt AI for the expected SP improvement. Source: Authors' own work

technology for hotels and found that PEOU is an aspect that impacts the adoption of the technology. [Guo et al. \(2023\)](#) examined how hospitality employees use TAM; this study will utilise meta-analytical structural equation modelling. The outcomes indicated that PU had a greater impact on “user attitudes” and “acceptance intention” than PEOU. Employees must believe that adopting AI in the hospitality sector will help sustain growth ([Agrawal et al., 2019](#)). Hence, the proposed hypothesis related to PEOU and PU is:

H1. PEOU will positively influence IAAI for the expected SP improvement.

2.3.2 Perceived usefulness (PU). PU refers to users' beliefs that utilising cutting-edge technology will improve their job performance within the firm. PU includes output quality, subjective norms, result demonstrability, and job relevance. Using AI in hospitality operations will help increase managers' job performance. [Rasheed et al. \(2024\)](#) found that PU is a primary driver in shaping consumers' intentions towards AI and robot adoption. AI adoption and PU can automate repetitive tasks while improving decision-making ([Rasheed et al., 2023a, b](#)). Better resource management and AI-powered predictive analytics might enhance performance ([Tavitiyaman et al., 2024](#)). AI adoption driven by high PU can improve customer satisfaction scores, operational efficiency, and profitability. For example, 24/7 customer service can be provided via a chatbot to reduce wait times, thereby improving the experience and the brand's reputation ([Huang et al., 2024](#)). [Hew et al. \(2018\)](#) measured tourists' mobile shopping intention and found a positive relationship between PU and their buying intentions. It will help in increasing customer satisfaction and customer presence.

H2. PU will positively impact IAAI for the expected SP improvement.

2.3.3 Government support (GS). Government policy and support are critical in stimulating IT innovation ([Chen et al., 2021](#)). As an innovative technology with many influences, AI raises several security, privacy, and social ethics issues. GS will create a positive atmosphere for adopting technology in the hospitality sector ([Brock and Von Wangenheim, 2019](#)). Various governments worldwide have invested vastly in technological adoption concerning AI ([Chen et al., 2021](#)). Establishing policymaking bodies to consider the societal, ethical, and legal implications is necessary for administration and business leaders to fully reap AI's rewards. [Yadegaridehkordi et al. \(2020\)](#) technologically advanced, a theoretical framework that includes the HOT and TOE frameworks to pinpoint the critical elements influencing the adoption of big data and its impact on firm performance. Generative storytelling is often the key catalyst of a technological adoption, especially in the hospitality industry. Having clear regulatory frameworks and guidelines to govern the use of AI will enhance compliance and risk mitigation and thus build confidence among the hotel operators. The technological infrastructure, especially broadband connectivity and data centres are part of the underpinning

that facilitates the implementation of AI-driven systems (Naz *et al.*, 2024). Nilashi *et al.* (2019) concluded that governmental policies are an indispensable component of developing medical tourism, as the example of the creation of the medical hotels in Malaysia shows.

H3. GS will positively impact IAAI for the expected SP improvement.

2.3.4 Competitive pressure (CP). CP refers to the business's intention to adopt the latest technology, taking into account market competition (Wu *et al.*, 2022). The IT revolution can bring many changes to firms by reshaping sector structures and finding innovative ways to compete in the market. The hospitality sector is under pressure if competitors are ready to adopt innovations in the market. King *et al.* (2020) address this scarcity by explaining how productive against unproductive worker voices can be realised. Chen *et al.* (2023) considered CP an essential factor for investigating AI adoption in the hotel sector. Kallmuenzer (2018) investigated which actors in the hospitality sector drive innovation in family-owned hospitality businesses, focussing on how these actors help establish a competitive advantage. Pizam *et al.* (2022) investigated the intention of hotel managers across 11 countries and found that CP is positively associated with the adoption of robotic technologies. CP in the market will help the employees improve their performance and achieve operational excellence compared to competitors in the hotel sector. As a result, the hospitality sector is compelled to adopt AI technologies due to CP. So, the proposed hypothesis is:

H4. CP will positively influence IAAI for the expected SP improvement.

2.3.5 Top management support (TMS). TMS mentions the higher authority's engagement in implementing the latest technology in the organisation. Hussein *et al.* (2019) showed a relationship between top management and PU for adopting B2B e-commerce in SMEs in Jordan. Higher management is essential for motivating employees to make sound decisions about technology adoption. Infante-Moro *et al.* (2021) identify the most critical factor influencing whether hotels accept and integrate IoT into their services. Leadership commitment significantly impacts adoption. TMS comes in many forms: financial support, motivating the managers to use the latest technology, training, and properly executing the implementation process. TMS can change the intention of adopting AI in the hospitality sector by explaining the benefits of AI implementation to managers (Enholm *et al.*, 2021). Hence, the proposed hypothesis is:

H5. TMS will positively influence IAAI for the expected improvement in SP.

2.3.6 IT infrastructure (ITI). ITI refers to the components required to operate and manage IT systems. ITI provides a competitive advantage and critical resources in any organisation (Toorajipour *et al.*, 2021). The components are the IT tangible resources, IT intangible resources, and IT intangible resources. A company's ITI is the support system for innovation and the adoption of new technologies. It is the basis for continuously incorporating IT innovations into businesses to boost performance. Today's global hospitality sector's performance depends on IT applications and high-quality infrastructure. ITI lays the groundwork for ongoing IT integration into business activities, thereby driving superior organisational performance. Wang *et al.* (2016) studied the intention to accept mobile reservation arrangements in the hospitality industry and highlighted the importance of ITI for adopting technology. Oikonomou *et al.* (2022) investigated the impact of ITI and employees' IT skills on business performance across the hospitality industry. AI adoption requires a developed and integrated ITI. The managers need knowledge and skills in IT (Ameen *et al.*, 2021).

H6. ITI will positively influence IAAI for the expected SP improvement.

2.3.7 Intention for the adoption of AI (IAAI) and SP. AI usage in the hotel industry is a major driving force of change in the sector. The growing need to rationalise the traditional operational processes with smart AI technology has triggered the designation of AI

technologies into the very channel of hospitality practice. The possibility of such integration brings in the aspect of improving the reputation of the industry, increasing the revenue levels and significantly improving the customer experience to new heights. [Chen \(2023\)](#) investigated how two determinants, namely, the predisposition of high-technology firm staff to use AI applications and the analytical power of big-data-based assets of firms, affect the operational performance. [Flavián et al. \(2022\)](#) investigated the impact of clients' technology willingness and service consciousness on their inclination to utilise AI in investment services. The hospitality sector needs to improve SP to increase customer footfall.

H7. Intention for the adoption of AI has a significant influence on SP.

3. Research methodology

This section presents the research methodology, including the research design, instruments, sampling techniques, data collection procedures, and the study sample size, as it examines AI adoption within India's luxury hospitality sector.

3.1 Survey instrument development

This study analyses the managers' intentions regarding the use of AI in luxury hotels. A luxury hotel offers luxurious accommodations for its guests. There are currently no standards for luxury hotels (e.g. star ratings). Hotels with four or five stars are considered "luxury." Additionally, this suggests that any accommodations can be considered luxurious ([Alsetoohy et al., 2019](#)). Most of the current questionnaire's items were adapted from previously established scales. The constructs PEOU and PU were adopted ([Davis, 1989](#)). The constructs TMS, ITI, GS and CP were adopted ([DePietro et al., 1990](#)). Academic and corporate experts helped to design the questionnaire (refer [Table A1](#) in [Appendix](#)). After preparing the questionnaire, it was sent to six academics and five industry experts in the hospitality sector for validation.

3.2 Research design, sampling and data collection

The present study focuses on managers working in luxury hotels across India that offer both lodging and dining facilities. This research uses an explanatory design since it intends to verify the hypotheses that are derived from theory and identify the causal relationships between the technological, organisational, and environmental factors influencing AI adoption through SEM. India may be one of the most topical locations to study how AI is adopted in the hospitality industry since the country is technologically developing rapidly, and the elements of digital transformation are getting more attention in the tourism and service sectors ([Ministry of Tourism, 2023](#)). The key aim of the research is to define and discuss the factors that affect the willingness of hotel managers to implement AI technologies in the operation of a luxury hotel. The research design that was used was quantitative, cross-sectional, to examine the interrelationships between the latent constructs developed in this study in a systematic manner, as suggested by [Kumar et al. \(2024\)](#).

In line with previous studies (e.g. [Chahal and Devi, 2024](#); [Douglas et al., 2026](#)), the unit of analysis was hotel managers who were directly involved in the strategic planning and the adoption of technologies. These interviewees were specifically chosen on the basis of their experience in operations, management supervision, and the level of understanding of AI-based applications in hospitality management. Only the respondents that proved to have enough knowledge on AI technologies and the ways of their use in business were included in the final sample to guarantee the relevance of the data and its accuracy.

Purposive sampling is the appropriate choice in the hotel industry as suggested in the recent research on the online hotels booking intentions of travellers by [Biswas \(2023\)](#). According to the previous literature the present study also uses purposive sampling method to determine the

appropriate respondents among luxury hotels who fit into the inclusion criteria of the study. The contact information of employees in hotels was mostly acquired through official websites of hotels, professional networking sites, and online communication. In other cases, the chat based inquiry services offered on the hotel websites were also used in an attempt to contact potential participants.

The Ministry of Tourism, Government of India classifies five-star deluxe, five-star, and four-star properties as a five-star hotel which is categorised as a luxurious hotel. In accordance with the study objectives, the term luxury hotels in this study is specifically used to denote five and four star hotels (Mohsin and Lockyer, 2008). Online and offline survey modes were also used so as to cover a wide geographical area and have more respondents in many regions of India.

The 21 luxury hotels in the 7 major Indian cities, namely Visakhapatnam, Hyderabad, Bhubaneswar (Odisha), Kolkata (West Bengal), Delhi, Mumbai, and Bengaluru gave the consent to data collection to the managerial staff. This regional representation also provided national coverage on eastern, southern and western regions. The researcher visited face-to-face hotels in Ahmedabad and Mumbai and received responses in the form of face-to-face interview whereas the other place received responses on Google Forms.

As emphasised by Bansal (2022), web-based surveys often surpass traditional postal surveys in terms of efficiency, accuracy, and response rates. Consistent with this evidence, the present study employed a Google Forms-based online questionnaire to maintain participant anonymity, complemented by in-person data collection to enhance representativeness.

The targeted population of this research was managerial-level employees who are working in four- and five-star luxury hotels located in metropolitan cities of India. The authors reached out to 1,475 hotel managers for the study. Out of them 1,123 were contacted through online invitations (email and digital platforms) and 352 were approached through personal visits. They got 312 complete and valid responses from the total pool of participants. Their overall response rate was 21.15%, and the non-response rate was 78.85%. The sample size they have attained is greater than the minimum SEM requirement suggested by Hair *et al.* (2014) which makes the statistical power for model estimation to be sufficient. Data collection was conducted over seven months, from May 2024 to November 2024. Before the primary survey, a pilot study with 95 respondents from various luxury hotels across India was conducted to test the questionnaire's reliability, validity, and clarity. The Cronbach's alpha (α) values for all constructs exceeded the 0.70 threshold, confirming strong internal consistency and measurement reliability (Hair *et al.*, 2014).

4. Data analysis and results

The rigour of the collected data was maintained throughout the analysis. After removing incomplete and extreme responses, 312 valid samples were analysed using SEM in AMOS 25.0 to examine the proposed relationships between the determinants of AI adoption in luxury hotels and SP. In this section, the study presents a common-method bias correction, a measurement model, and a structural model for path analysis, along with empirical validation.

4.1 Demographics

The demographic profile of 312 respondents illustrates a varied and skilled managerial workforce of India's luxury hotel sector. The sample evidences gender distribution without bias and comprises the representatives of the managerial personnel, where IT managers, being the largest group, have most of the influence. Around 80% of the respondents are aged 26–35 years, thus suggesting a relatively young demographic, which is still professionally active. Over half of them are postgraduates, thus pointing to their strong educational background. The respondents' industry experience is of great length as more than 75% of them have got over six years of the professional exposure. Consequently, the demographics point to

a well-educated and experienced pool of respondents which is appropriate for the AI adoption evaluation in the hospitality industry.

4.2 Reliability and validity

Common method bias was tested to ensure that the collected data were free from bias. Second, we calculated reliability using Cronbach's alpha. Cronbach's alpha measures data reliability (Hair *et al.*, 2012). Table A2 (Appendix) shows that Cronbach's alpha values for the items shown are greater than or equal to 0.70. Common-method bias (CMB) occurs when respondents' true predispositions lead to discrepancies between the instrument's responses and the information it seeks to elucidate. Harman's single-factor test was used after the data collection. Finally, the results show that the 1st component accounts for the majority of the variance or covariance, up to 15.867%, which is below the advised level of 50% (Podsakoff, 2003).

All the factor loadings are found to be greater than or equal to 0.5, which is the acceptance level. Table A2 (Appendix) shows the factor loadings. This study employed CB-SEM, which is suitable for factor-based models, and confirmed reliability using Cronbach's alpha (0.800–0.904). Convergent validity was established by AVEs and CRs exceeding 0.5 and 0.7, respectively. Discriminant validity was demonstrated using Fornell-Larcker criteria and HTMT ratios below 0.85. Finally, the model exhibited good fit indices: CMIN/df = 1.567; CFI = 0.955; TLI = 0.951; IFI = 0.956; NFI = 0.886; and RMSEA = 0.043, confirming the conceptual framework's validity and reliability. The composite reliability values are more significant than 0.70, indicating the constructs' good indicator reliability (DeVellis and Thorpe, 2021). This study found that all the AVE values were more significant than 0.50 (Fornell and Larcker, 1981). After evaluation, it was found that the construct pairs demonstrated discriminant validity. The results are shown in Table A3 (Appendix).

4.3 Results of hypotheses

The structural equation modelling technique used for hypothesis testing is presented in Figure A1. Table 1 indicates that one hypothesis was not supported, whereas the other six hypotheses were supported.

The current study examines hotel managers' intention to implement AI to enhance guest satisfaction by systematically analysing the factors influencing AI adoption through the integration of the TAM and TOE frameworks. Roy *et al.* (2020) noted that Indian customers are involved in the implementation of AI-based technologies. These technologies can increase the customer desire to visit the hospitality industry. Respectively, the knowledge produced in this study may be useful to the hospitality sector to improve customer experience, thus, possibly increasing profit margins. Based on this, companies have shifted their strategic focus towards survival in these new challenges.

Table 1. Path analysis results for the hypothesis

Parameters	Estimate	SE	CR	<i>p</i>	Remarks
PEOU → Intention for the adoption of AI	0.128	0.050	2.121	0.034	Supported
PU → Intention for the adoption of AI	−0.003	0.054	−0.049	0.961	Not Supported
TMS → Intention for the adoption of AI	−0.004	0.055	−0.075	0.940	Not Supported
ITI → Intention for the adoption of AI	−0.142	0.040	−2.331	0.020	Supported
CP → Intention for the adoption of AI	0.272	0.060	4.457	***	Supported
GS → Intention for the adoption of AI	0.223	0.044	3.786	***	Supported
Intention for Adoption of AI → SP	0.346	0.060	5.083	***	Supported

Source(s): Authors' own work (Generated through software -AMOS 25.0)

The adoption of AI by the competitors can also contribute to the increase of market saturation and a stronger competition, thus making firms that will not use the technologies disadvantaged. This empirical study looked at the experiences of hotel employees using AI hence disclosing potential areas of challenge as well as opportunities to service organisations. The results also helped to explain the convergent effect of factors, such as PEOU, PU, managerial support, infrastructural readiness, CP, and policy considerations, on AI acceptance. The new study should also consider these results and apply them in various cultural and geographic settings to further advance the general comprehension of AI usage in the hospitality industry.

Hypothesis (H1) tested the relationship between PEOU and IAAI regarding the expected SP improvement, and the hypothesis was supported. AI-based technologies are faster in solving customers' problems. Managers believe their jobs will grow in value. (Chatterjee *et al.*, 2021) detected that PEOU and PU positively influence the intention of AI adoption. (Rasheed *et al.*, 2023a, b) performed semi-structured interviews in restaurants in Pakistan and found that people preferred to go to hotels that use technologies like robots, AI, etc. Also, respondents said that they would choose AI in hotels because it is easier to order and deliver food. PEOU was found to have a positive association between humans and robots during the implementation of various operations. This study found that managers believe their systems will become effort-free with AI-based technologies in the hospitality sector. (Pillai and Sivathanu, 2020a, b) interviewed hotel managers adopting AI-based chatbots, and the interviewee noted that most customers own cell phones. (Rasheed *et al.*, 2023a, b) interviewed a few customers in Pakistan for AI-based robots in the hotel industry, where the interviewees confirmed that they prefer to go for dinner in those hotels which use AI-based technologies, as it will have an easy handling process; the interviewee prefers that the robots serve the food; the interviewee feels that AI-based robots will perform better and efficiently in the hotels.

The second hypothesis showed an unsupported relationship. (Tavitiyaman *et al.*, 2022) show that PU is strongly associated with the intent to adopt technologies among hospitality customers. Cai *et al.* (2022) found CP to influence consumers' intent to utilise AI-enabled voice assistants. Pillai and Sivathanu (2020a, b) found that positive relates to adopting AI-based chatbots. The study by Rasheed *et al.* (2023a, b) shows that people judge AI-based robots' performance better than humans. Also, people responded that they and their families want to go to that type of restaurant with AI-based robots to improve time efficiency. Hypothesis (H3) tested the relationship between GS and IAAI regarding the expected SP improvement, and the hypothesis was supported. GS was found to be significant in previous studies of website adoption in micro-enterprises (Osakwe *et al.*, 2016). The hospitality sector wants the government to provide benefits for technology adoption. The rules and regulations for technology adoption need to be appropriately framed. Still, hotels must have some restrictions when dealing with customers.

This study confirmed that CP positively influences IAAI (H4) for service improvement, suggesting that CP is not a significant barrier in hospitality; however, (Nam *et al.*, 2021) The study found that managers acknowledge a desire to follow successful AI trends, but they may not perceive immediate competitive threats locally. Hypothesis (H5) tested the relationship between TMS and IAAI regarding the expected SP improvement, but this relationship was unsupported. (Alsetoohy *et al.*, 2019) found that the TMS of hotels does not support the adoption of technological innovations. Further, TMS is interested in adopting AI-based skills in the hospitality sector. They are ready to invest and provide proper training to their managers for the adoption process. Hypothesis (H6) tested the relationship between ITI and IAAI regarding the expected SP improvement, and the hypothesis was supported. (Pillai *et al.*, 2021) found that the ITI in auto components manufacturing companies is insufficient to implement AI. (Wei *et al.*, 2015) found that ITI is sufficient for implementing RFID in Chinese firms. A lack of ITI can lead to massive investment by firms in the adoption process. This study found that the hospitality sector does not need to invest in adopting AI-based technologies. Hypothesis (H7) tested the relationship between IAAI and SP, and the hypothesis was

supported. SP improvement can help drive the growth of the hospitality industry and create competitive advantages in the market. SP can help increase customer satisfaction. AI and AI-based technologies can help improve SP more competently (Cai *et al.*, 2022).

Bounatirou and Lim (2020) selected a case study of a hotel chain based in the United Kingdom and proved that AI can positively affect the quality of services and business performance in the hospitality industry. Lukanova and Ilieva (2019) described the implementation of robotics, AI, and service automation in five different stages, namely, pre-arrival, arrival, stay, departure, and assessment. Therefore, data collection and reservation processes can be utilised in pre-arrival AI systems. Once the guest arrives at the hotel, the hotel robots undertake reception and registration of the guest and also assign the room to that guest. Furthermore, new hotel facilities have also integrated AI to collect the individual preferences of clients, which allows management to create personalised services that will add to the guests. Automation of the service at this point saves on time and effort by the patrons, and thus enhances their positive attitudes towards the business premise and the overall customer experience (Gajić *et al.*, 2024). Data analytics also offers high potential with the implementation of AI. Customer reviews can be tracked and analysed by AI platforms on multiple platforms, such as booking transactions, satisfaction surveys, and third-party platforms (Chen *et al.*, 2024; Huang *et al.*, 2024). The major international hotel brands like Caesars and Hilton are using AI to manage rooms, make reservations, and provide personalised services, and many others, including FlyZoo, are using facial-recognition technology. AI also streamlines occupancy, utilisation of resources, and revenue, which makes the operations run efficiently and focus on the target customer (Tung and Au, 2018).

5. Implications of the study

5.1 Theoretical/academic implications

This study makes several meaningful theoretical contributions to the existing literature on technology adoption, service innovation, and hospitality management by extending, refining, and applying established theories in relevant contexts. First, it refines the application of the TAM (Davis, 1989) and the TOE framework (DePietro *et al.*, 1990) by showing the merits of a combined approach. While TAM has conventionally focused on individual-level belief factors, such as PEOU or PU, and TOE stresses organisational and environmental factors, this study combines both theories to provide a more comprehensive view of organisational decision support adoption within organisational decision support processes. This dual-theory approach allows for a deeper examination of the interplay between cognitive perceptions and institutional pressure, managerial support, and technological capabilities, which, according to earlier research, have often been examined independently (Chatterjee *et al.*, 2021; Nam *et al.*, 2021).

Second, the paper contributes to the contextual extension of adoption theories by investigating AI adoption in an emerging economy setting. Although the majority of earlier studies have drawn on developed markets (Belanche *et al.*, 2021; Tung and Au, 2018), our results demonstrate that environmental factors, for instance, governmental support and competition pressure, have a more decisive impact on managerial adoption decisions based on developing markets, for which institutional maturity and information infrastructure are continuously evolving. This contributes to the widening of the external validity of TAM–TOE-based models, allowing for theoretical refinement of adoption behaviour based on the shaping impact of setting contexts. Third, this study broadens the end-focused dimension of the literature on technology adoption by relating the intention to adopt AI to SP. Most prior work has confined its scope primarily to measuring adoption intentions, without examining subsequent effects (Seo and Lee, 2021; Pillai and Sivathanu, 2020a, b). This study demonstrates that adopting AI significantly enhances SP, reflected in service quality advancements, personalisation, responsiveness, and efficiency, thereby bridging the gap between adoption decisions and organisational outcomes. This finding adds a thread to the

theoretical discussion by positioning SP as a crucial axis in the examination of technology adoption in service industries.

Finally, the work contributes to theory development by adding new knowledge about the relationships between constructs. The absence of significance for PU, on top of the undesirable correlation between ITI and adoption intention, suggests that, in early adoption situations, ease of use, pressure from competitors, or governmental support may outweigh usefulness perceptions or infrastructure readiness. This goes against received knowledge about TAM, yet it provides a conceptual exploration of situational-specific moderation effects.

5.2 Policy and managerial implications

This study proved that managers were ready to adopt AI-based technologies for sustainable growth. This study will provide favourable conditions for adopting AI in the business sector. It also highlights the hospitality sector's readiness for AI adoption. It was found that the hospitality sector has the adequate ITI required for AI adoption. AI-based technologies are more efficient at resolving customer issues. Managers also trust that AI-based technologies will improve performance and productivity, enabling sustainable growth. Managers believe that their jobs will pay off financially in the future. Hotel managers must provide every detail of technology adoption to their employees and stakeholders. Managers must make the stakeholders understand the importance of AI and its technologies. The use of AI-based technology requires proper knowledge and training. Adequate information and training will enhance employees' intention to adopt AI. ITI plays an essential role in knowledge adoption. So, the managers must ensure that sufficient infrastructure is available for AI adoption.

Managers need to ensure that hotels are ready for AI adoption by developing effective strategies and planning. Managers must maintain appropriate communication among all employees regarding the technology adoption process and its related information. Managers must make employees understand that AI adoption will increase job performance and sustainable hospitality growth. Managers need to create a positive environment for customers to adopt AI. Some of the recommendations and suggestions from the managers during the study could be helpful for the AI adoption process. Awareness of contactless food delivery needs to be increased among customers. This contactless delivery of the restaurant's food will also help prevent the panic caused by the pandemic. Customers will not fear COVID-19 virus transmission. This type of contactless food delivery can be achieved with AI-powered robots. Employees need to highlight the positive aspects of AI adoption to avoid resistance to change. The implementation of AI necessitates governmental assistance. The hotel sector must actively push AI systems or AI-powered robots to attract clients. The hospitality sector must implement AI-powered systems to minimise food waste and sustain operations across SC. The hospitality sector needs to adopt green technologies that can reduce environmental emissions. The hospitality sector must adopt water and waste management technologies for environmental sustainability.

The enhancement of SP and the application of AI in the hospitality industry should be placed in the context of a macroscopic framework, which will be more focused on sustainability, climate action, and inclusive development. Examples of such technological effects of AI can be demonstrated through empirical case studies that the technology impacts can be leveraged to support these agendas. An example is the case of community-based tourism in East Africa, where digital platforms and AI-based tools based on demand forecasting have been used to control visitor arrivals according to ecological conservation goals, thus improving the environmental and economic stability of the local community. Southeast Asian eco-lodge parallels have used AI-based resource-management and predictive maintenance platforms to reduce water and energy use and at the same time enhance guest experience, thus pointing to the reality of technological adoption as sustainable tourism and climate change. These practical findings indicate that hotel administrators and policy-makers need to understand the introduction of AI as a vital component of an overall sustainability

transition plan and not a mere technological addition. However, the integration of AI into a wider range of programmes aimed at sustainability, including waste management, energy efficiency, and community-based programmes, might not result in the enhancement of the performance of the services, nor their capacity to support the sustainability outcomes over time, destination resilience, and the adherence to global sustainability targets, including the Sustainable Development Goals.

5.3 Practical implications

The results of this research make a number of substantive pragmatic contributions on practitioners in the hospitality and tourism industry, policy makers, and industry stakeholders in the value chain. To begin with, they argue that the adoption of AI is not an act of technical supplementation but a strategic pathway to achieving service predominance, strengthening the business, and securing competitive differentiation. When hospitality enterprises plan to implement AI at every stage of their value chains, they will have an opportunity to afford hyper-personalised guest experiences, increase service quality and responsiveness, and improve the work efficiency. These advantages could be achieved by implementing AI-powered chatbots to provide 24/7 communication with guests, using predictive analytics to predict demand, and implementing robotic automation in order to optimise housekeeping services and front-office operations.

Second, the paper highlights the importance of capacity building and redesigning of the workforce. The implementation of the AI requires the retraining of staff, the development of online skills, and the adoption of the culture of decision-making based on data to stimulate innovative working practices. Training and partnerships with information-technology suppliers can provide the existing gaps in digital literacy and enhance the ability of the workforce to process, analyse, and operationalise insight based on AI systems with effectiveness.

Third, the study provides practical recommendations to policy making and planning of the industry. Governments are being requested to implement adequate regulatory mechanisms, invest and provide funding opportunities and infrastructural development ventures that will encourage AI utilisation, especially among small and medium-sized hospitality businesses. Also, the combination of AI and green projects, including energy efficiency and waste reduction, and supply-chain transparency has potential to support the development of environmental transformation goals and meanwhile improve the efficiency of the economy. Lastly, the research outlines how innovation ecosystems between technology companies, academic institutions, and tourism agencies become key to the scale-up of AI solutions in the hospitality industry. These platforms have the potential to hasten the spread of innovation and capitalise on its social, economic, and environmental gains and use them as the foundation of overall developmental returns.

5.4 Technological implications

The research also provides key technological contributions by demonstrating AI's capacity to serve as a transformative agent for service development and operational policies. Firstly, it is discovered that the hospitality industry's tech adoption needs to move beyond incremental automation to embrace intelligent, whole-system approaches that adaptively shift to customers' evolving requirements and market dynamics. With technologies like machine learning, natural language processing, and AI-driven computer vision, it is possible to achieve predictive maintenance, demand forecasting, focus-based marketing, and real-time decision-making, thereby transforming conventional service processes into intelligent, data-driven systems.

Thirdly, it also seems, based on the study's findings, that its interoperability and technological harmonisation for a successful implementation of AI. It's of major importance for hospitality companies to verify whether their AI technologies are harmonised with their

current property management systems, customer relationship management programs, and Internet of Things (IoT) devices to improve SP and influence maximisation. Moreover, AI integration of complementary technologies, e.g. cloud computing, robotics, and blockchain, may significantly strengthen service operations' transparency, efficiency, and scalability.

Third, it highlights the importance of continuous technological development and supporting infrastructure development. As AI advances, future hotel systems are expected to incorporate self-adaptive algorithms, edge computing, and generative AI to deliver highly customised, context-sensitive, and self-sufficient services. Such technologies are going to not only transform guest experiences but also empower hotels to dynamically adapt to disruptions, optimise resource allocation, and build intelligent, self-learning service environments. It will be important for professionals in the industry to adopt a forward-looking strategy for responding to these technological advances to preserve a competitive edge and achieve sustainable digital transformation across the hospitality industry.

5.5 Limitations of the study

Although the research has been helpful, it still has some limitations. For example, the research is only about the managers of luxury hotels in India, so the results might not be the same for non-luxury hotels or other places. Also, the study didn't take into account the size of the organisation, how digitally advanced it is, or the manager's experience, and this could affect the results.

Another limitation of this study is that it does not have control variables in the conceptual framework. These include organisational size, digital maturity level, technological experience of managers, and type of hotel (4-star vs 5-star), which may affect the decisions of AI adoption. Omitting these controls may introduce omitted variable bias in the model estimates. Subsequent studies should incorporate appropriate control variables not only to improve the reliability of AI adoption models but also to enhance their explanatory capacity in the hospitality sector. The survey was done at one point in time, so it doesn't show how the use of AI has changed over time.

6. Conclusions and future research

The purpose of the proposed study was to uncover the factors that determine the use of AI in the hospitality industry and to evaluate its effect on the service performance by combining the TAM and TOE frameworks. The qualitative, hypothesis-oriented methodology was used whereby data were gathered in relation to 312 managers in luxury hotels in major Indian cities and examined through SEM. The research gives an in-depth insight into the ways technological, organisational, and environmental factors interact to predict managers' and their intention to use AI and performance.

The results show that PEOU, CP, GS, and ITI have a major effect on the intention to use AI, and perceived usefulness and top management support did not prove to be of great importance in this situation. Moreover, the use of AI shows a significant benefit of a positive impact on the service performance, increasing efficiency, responsiveness, and customer satisfaction. These findings emphasise the increasing strategic value of AI-based technologies in enhancing operational efficiency and providing a high-quality service experience in the hospitality sector.

The research will add value to the theory by incorporating TAM and TOE as one framework, which provides a more comprehensive picture of the organisational technology adoption in the context of an emerging economy. It is also built upon the literature, as it connects AI adoption intention to real service outcome performance. In practical terms, the results can inform the hotel managers and policymakers to focus on easy-to-use AI systems, enhance digital infrastructure, and use external pressures and policy facilitation to hasten its adoption. In general, the paper highlights AI as a key facilitator of service quality and competitiveness in changing hospitality industry.

The researchers may broaden the spectrum of the hospitality businesses to include boutique hotels, budget hotels, and restaurants for the generalisability of their findings. They may as well include some relevant control variables to depict the characteristics of an organisation and lessen the possible bias of the model. Moreover, customer acceptance of AI technology in services is an alternative angle to adoption and SP which future studies can consider. Lastly, the exploration of the combination of AI with other emerging technologies such as blockchain, big data analytics, and robotics would reveal more about the digital transformation process in the hospitality sector. Researchers in the future can use a longitudinal design to see changes in adoption behaviour and SP outcomes over time.

Data availability

The data will be provided upon request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT to improve the language of the work. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Appendix

Table A1. Measurement items and sources

Latest variables	Indicators	Source
Perceived ease of use (PEOU)	PEOU 1: We can understand the process of AI and its usage PEOU 2: Our hotel can efficiently operate AI-based systems PEOU 3: We are confident in using AI systems in my hotel PEOU 4: We are confident that all our hotel employees can easily and speedily learn about AI usage	Davis (1989)
Perceived usefulness (PU)	PU 1: Our hotel will become more efficient using AI PU 2: We trust an AI-based system will increase the hotel's productivity PU 3: We can quickly achieve things using an AI-based system PU 4: AI-based systems will reduce costs	Davis (1989)
Top management support (TMS)	TMS 1: Top authorities are very much supportive of implementing AI-based systems TMS 2: Top management motivates managers in support of AI TMS 3: Top authorities do not support the adoption of technology TMS 4: Top management will provide all the essential AI adoption resources	DePietro <i>et al.</i> (1990)
IT infrastructure (ITI)	ITI 1: The IT infrastructure required for AI-based technology is available in our hotel ITI 2: Our managers are technically skilled in handling AI-based systems ITI 3: Our hotel's employees and staff want to learn AI-based technologies	DePietro <i>et al.</i> (1990)

(continued)

Table A1. Continued

Latest variables	Indicators	Source
Competitive pressure (CP)	CP 1: There has been an increase in technology adoption across the hotel sector CP 2: Only the latest innovative technologies can help our hotel serve the customer better CP 3: There is pressure from the competitors as it uses the latest innovative technology	DePietro <i>et al.</i> (1990)
Government support (GS)	GS 1: Government support in the technology adoption process GS 2: Government policies towards AI will be helpful in the adoption of AI by hotels GS 3: The support from the government is secondary to the adoption of AI in my hotel	DePietro <i>et al.</i> (1990)
Intention for the adoption of AI (IAAI)	IAAI 1: We intend to use AI in the future IAAI 2: We will continuously try to use AI in my workplace IAAI 3: We plan to use AI frequently IAAI 4: We think using AI-based technology will enhance our hotel productivity	Chatterjee <i>et al.</i> (2021)
Service performance (SP)	SP 1: AI adoption will increase the hotel's service performance SP 2: Customers will be more satisfied with improving service performance SP 3: There will be more and more customer footfall with improving service performance using AI SP 4: There is no chance of improvement in service performance using AI	Cai <i>et al.</i> (2022)
Source(s): Authors' own work		

Table A2. Values of Cronbach's alpha, AVE, CR and factor loadings

Latent variables	Cronbach's alpha	Factor loadings	AVE	CR
Perceived ease of use (PEOU)	0.8	0.786 0.846 0.856 0.586	0.526	0.808
Perceived usefulness (PU)	0.865	0.849 0.840 0.837 0.811	0.617	0.865
Top management support (TMS)	0.901	0.848 0.883 0.900 0.864	0.696	0.901
IT infrastructure (ITI)	0.837	0.879 0.881 0.822	0.636	0.839
Competitive pressure (CP)	0.878	0.843 0.898 0.893	0.71	0.88
(continued)				

Table A2. Continued

Latent variables	Cronbach's alpha	Factor loadings	AVE	CR
Government support (GS)	0.897	0.911 0.904 0.858	0.753	0.901
Intention for adoption of AI (IAAI)	0.904	0.856 0.841 0.860	0.702	0.904
Service performance (SP)	0.807	0.864 0.743 0.838 0.830 0.718	0.811	0.52

Source(s): Authors' own work**Table A3.** HTMT test and Discriminant validity

	IAAI	PEOU	TMS	PU	SP	GS	ITI	CP
<i>Discriminatory analysis</i>								
IAAI	0.838							
PEOU	0.057	0.725						
TMS	0.025	0.189**	0.834					
PU	0.015	0.275***	0.097	0.785				
SP	0.349***	0.054	0.082	0.116†	0.721			
GS	0.262***	-0.248***	0.018	-0.011	0.038	0.868		
ITI	-0.164*	0.084	-0.052	-0.052	0.004	-0.09	0.797	
CP	0.318***	-0.033	0.005	-0.052	0.164*	0.231***	-0.053	0.843
<i>HTMT test</i>								
IAAI								
PEOU	0.059							
TMS	0.024	0.187						
PU	0.021	0.313	0.1					
SP	0.365	0.033	0.083	0.11				
GS	0.262	0.24	0.004	0.003	0.051			
ITI	0.179	0.072	0.036	0.048	0.009	0.099		
CP	0.322	0.042	0.015	0.042	0.172	0.253	0.074	

Note(s): Significance of Correlations: † $p < 0.100$, * $p < 0.050$, ** $p < 0.010$, *** $p < 0.001$ **Source(s):** Authors' own work

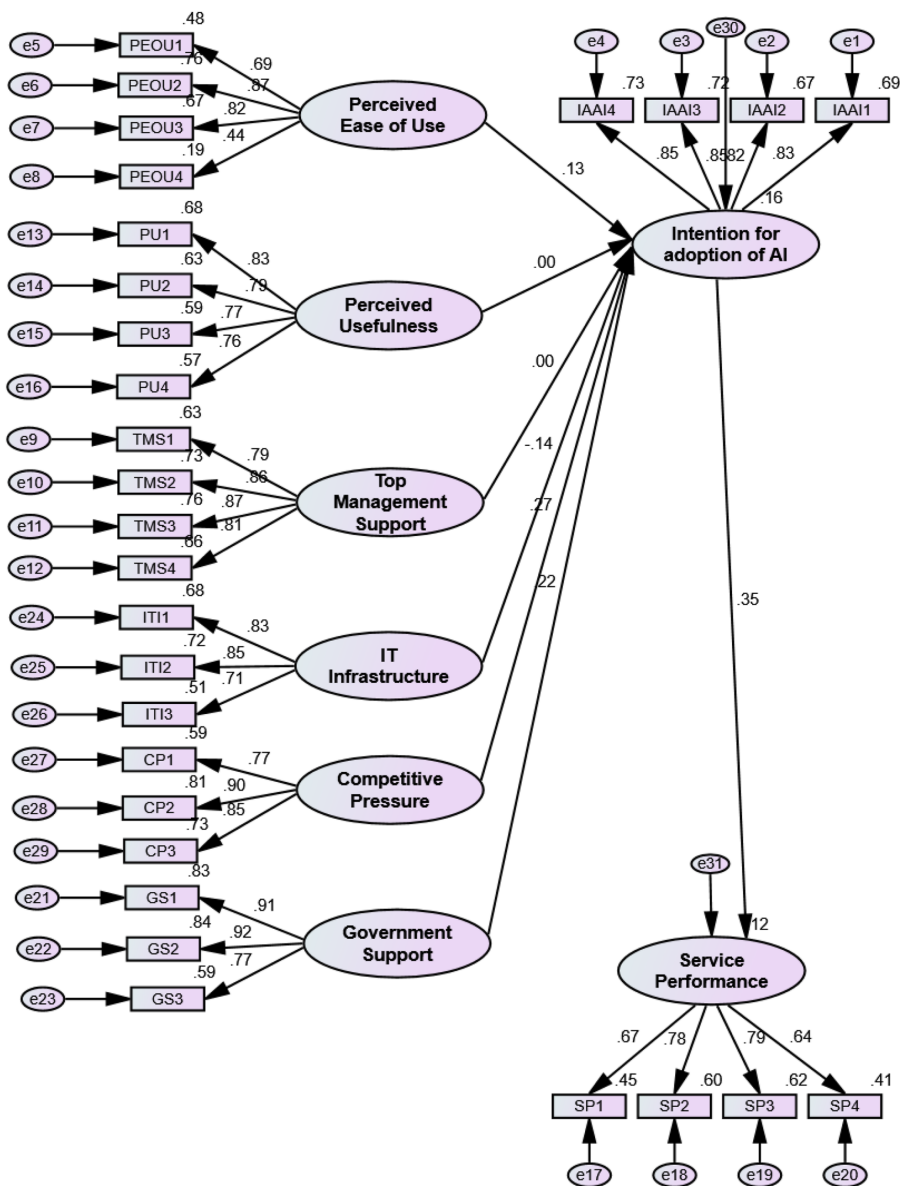


Figure A1. Structural model for Intention of AI adoption. Source: Authors' own work

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